THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A letter is chosen at random from the word 'ASSASSINATION'. The probability that it is the letter S is: 1

(a) $\frac{3}{13}$ **(b)** $\frac{4}{13}$ **(c)** $\frac{9}{13}$ **(d)** $\frac{5}{13}$

2. A bag contains 6 red and 4 blue balls. 4 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{5}{7}$ **(b)** $\frac{9}{14}$ **(c)** $\frac{11}{14}$ **(d)** $\frac{2}{7}$

3. A box contains 24 tickets numbered 1 to 24. One ticket is drawn at random. The probability that the number drawn is divisible by both 2 and 3 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{5}{24}$

4. A lot of 200 ball pens contains 24 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{177}{200}$ **(b)** $\frac{3}{25}$ **(c)** $\frac{22}{25}$ **(d)** $\frac{7}{8}$

5. Two dice are thrown simultaneously. The probability that the sum is at least 10 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{7}{36}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{1}{6}$

6. A coin is tossed three times. The probability of getting exactly one head is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

7. A letter is chosen at random from the word 'ELEVATION'. The probability that it is the letter E is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{9}$

8. A bag contains 2 red and 8 blue balls. 3 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{6}{13}$ **(b)** $\frac{5}{13}$ **(c)** $\frac{4}{13}$ **(d)** $\frac{8}{13}$

9. A box contains 50 tickets numbered 1 to 50. One ticket is drawn at random. The probability that the number drawn is a number divisible by 5 is: 1

(a) $\frac{9}{50}$ **(b)** $\frac{11}{50}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{1}{5}$

10. A lot of 160 ball pens contains 8 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{151}{160}$ **(b)** $\frac{153}{160}$ **(c)** $\frac{1}{20}$ **(d)** $\frac{19}{20}$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A letter is chosen at random from the word 'COMMITTEE'. Find the probability that it is the letter M. **2**

12. A coin is tossed three times. Find the probability of getting at most two heads. **2**

13. A red die and a blue die are rolled together. Find the probability that the sum is a prime number. **2**

14. A letter is chosen at random from the word 'TRIGONOMETRY'. Find the probability that it is a consonant. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two dice are thrown simultaneously. Find the probability that the sum is 5. **3**

16. A box contains 32 tickets numbered 1 to 32. One ticket is drawn at random. Find the probability that the number drawn is a perfect square. **3**

17. A bag contains 5 red and 10 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. **3**

OR

Two dice are thrown simultaneously. Find the probability that the sum is 9. **3**

18. A box contains 35 tickets numbered 1 to 35. One ticket is drawn at random. Find the probability that the number drawn is a number divisible by 5. **3**

OR

A bag contains 9 red and 3 blue balls. 6 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): **5**
(i) the sum is 12,
(ii) the sum is 5,
(iii) the sum is 9 (in this order).

OR

A bag contains 2 red and 10 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{3}{8}$. Find the value of x. **5**

20. Two dice are thrown together. Find the probabilities of these events (in this order): **5**
(i) both numbers are even,
(ii) at least one die shows 6,
(iii) the sum is 3 or 11 (in this order).

OR

A bag contains 7 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{16}{21}$. Find the value of x. **5**

END OF QUESTION PAPER — Total: 40 Marks